

Static Gesture Classification — Run Report

Generated 2026-05-10 22:02:42

Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 30 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 30 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 30 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 30 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 30 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 30 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 30 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 31 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 30 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 31 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 30 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 32 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 30 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Static · 0 trials
Training data path 16	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 0 trials
Total training paths	16 · 424 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 9504
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=339 · test=85

Augmentation (training only)

Enabled	no
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Voting Soft (SVM+RF+KNN+LogReg)	0.9793	0.0189	1.0000
Stacking (SVM+RF+KNN+LogReg)	0.9881	0.0145	1.0000
SVM (Linear)	0.9793	0.0189	0.9882
Random Forest	0.9941	0.0118	0.9882
Logistic Regression	0.9705	0.0263	0.9882
SVM (RBF)	0.9704	0.0265	0.9765
LDA	0.9117	0.0557	0.9412
KNN	0.7670	0.0199	0.8353

Per-class hold-out F1

[illegible]

